

MAVRA learning, optimization and maintenance

1-2

learning

3

optimization

4-7

maintenance

other c

relearn

1 Learn

built-in agent (LLM)

question: March store revenue
SUM(ss_net_paid) ... d_moy = 3
answer 13,570,368.70, verified
validated before publishing



2 Publish v1

the definition holds

SUM(ss_net_paid)
rules: one row per sale,
date key unique,
loss in bound; learning SQL

3 Optimize

period predicate as a key range

14 periods agree ✓
29.1 → 23.3 ms, -19.8% ✓
new condition: contiguous keys
publish v2; v1 still accepted

✗ No repair

e.g. a batch loaded twice

one row per sale: fails ✗
no filter restores it
retired; agents told
retired until relearned

4 Restatement

old rows kept as non-current

store_sales 13 → 14
one row per sale: fails ✗
v2 retired
unfixed: old rows counted

a later write

5 Find a repair

try filters on few-value columns

ss_is_current='1' ✓
ss_is_current='0' loses sales ✗
exactly one works ✓

6 Regression test

rerun March on learning-time data

learning SQL 13,570,368.70
with the filter 13,570,368.70
same: fix accepted ✓
current data would refuse it

7 Publish v3

new version, same metric

v3 = v2 + filter
ss_is_current='1'
SQL naming v2: rejected
waiting requests get v3

none (or several) work